STREAMLIT:

The fastest way to build Data apps

Steven Kolawole

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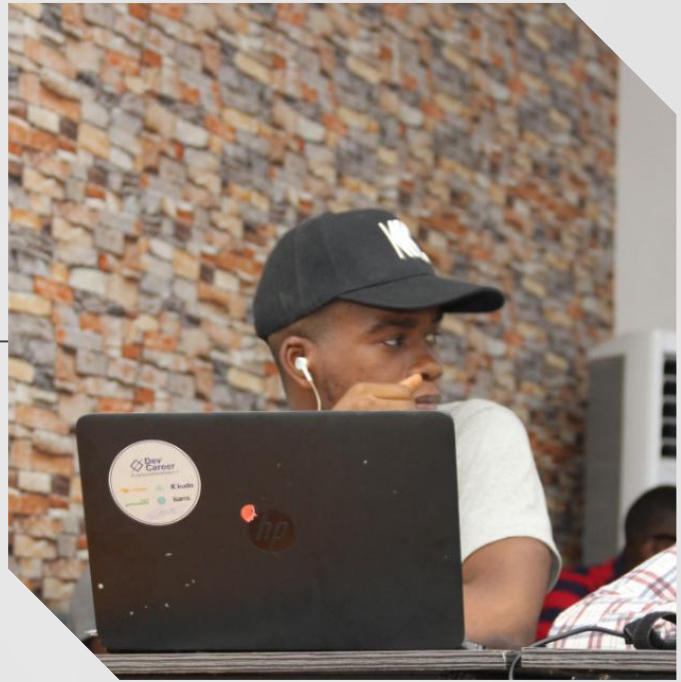
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ABOUT ME

Steven Kolawole

- An ML Research Engineer **looking for his next gig**
- Somewhere in the intersection between Software Engineering & Machine Learning
- Liverpool FC fan since kid
- Python Freak
- Foodie





01

Problem Statement

Building a Data-driven Web App Process



Data processing and analysis in notebooks



Copy-paste in a .py script



Create an app via Flask, and at least, HTML, CSS, JavaScript and JSON



Outlining requirements and UI/UX designing



Create a state-of-the-art app



Updates once in 3 months or so

The Pain Points

- Depending on the software engineering team takes up a bunch of our time
- We can't create new features and updates because the software dev team have lots on its plate

We'd create those apps ourselves but;

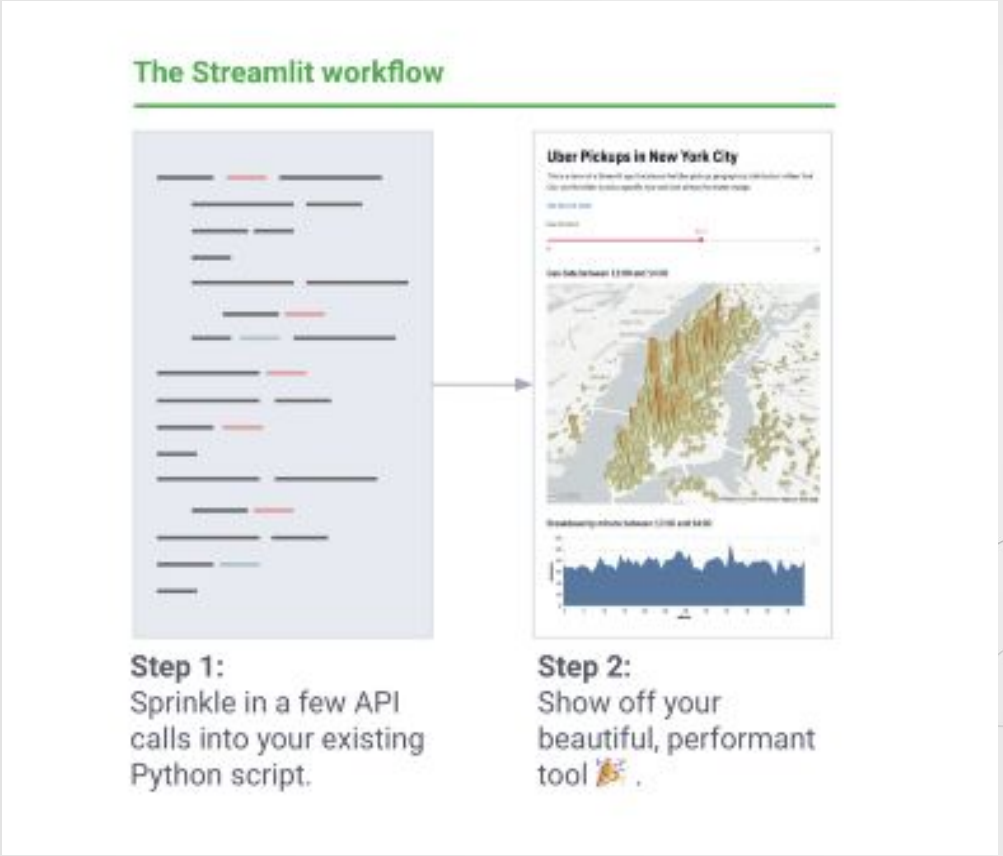
- We are not particularly artsy, and we suck at UI designs
- Creating web apps consume a humongous amount of time
- Aside from Flask, we have to bother with JS, HTML, CSS, JSON, HTTP requests and others
- We prefer to use code to generate insights or build a model, than providing user with aesthetic experience.

What if we could build apps as easily as writing Python Scripts???

02

What is Streamlit?

- Building apps as easily as writing Python scripts



- It covers most UIs used in data apps

```
def add_boxes(image, boxes):
    image = image.astype(np.float64)
    for _, (xmin, ymin, xmax, ymax, label) in boxes.iterrows():
        image[ymin:ymax,xmin:xmax,:] += LABEL_COLORS[label]
        image[ymin:ymax,xmin:xmax,:] /= 2
    return image.astype(np.uint8)

def main():
    st.write("# Let's Download Some Data")

    metadata = load_metadata(LABELS_FILENAME)

    # st.image(image, use_column_width=True)

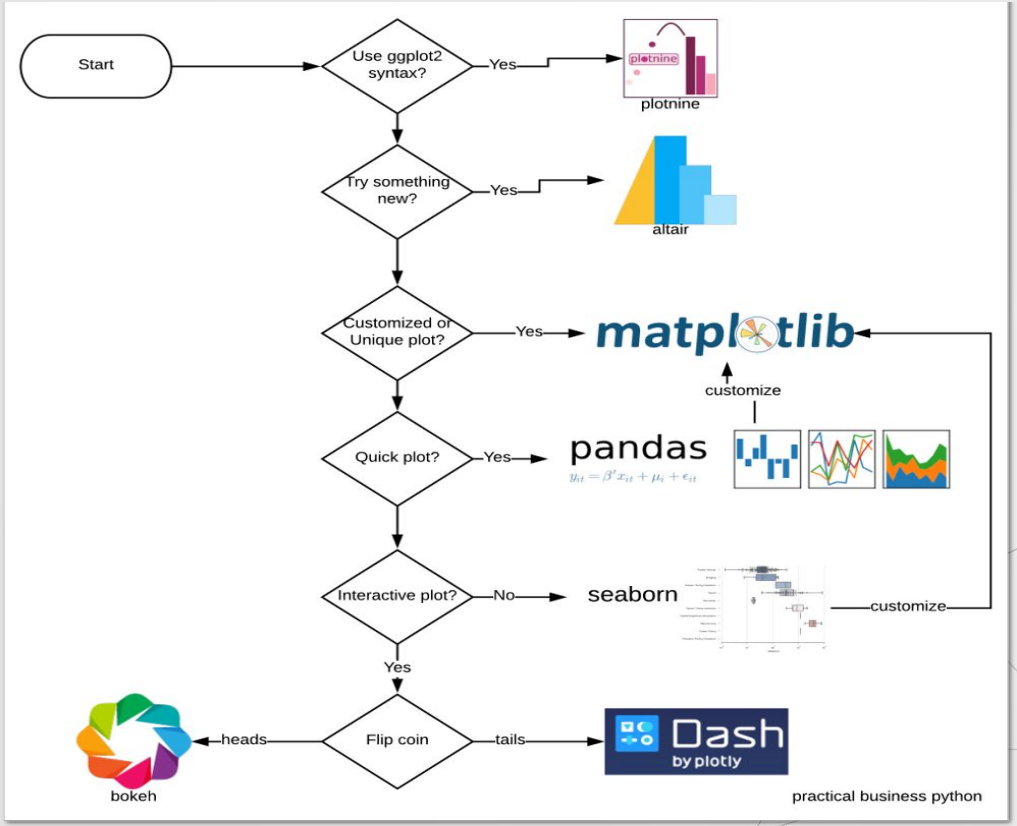



if __name__ == '__main__':
    main()
```

Streamlit

Let's Download Some Data

- Supports multiple interactive visualization libraries like ploty, altair, bokeh, pydeck, etc.

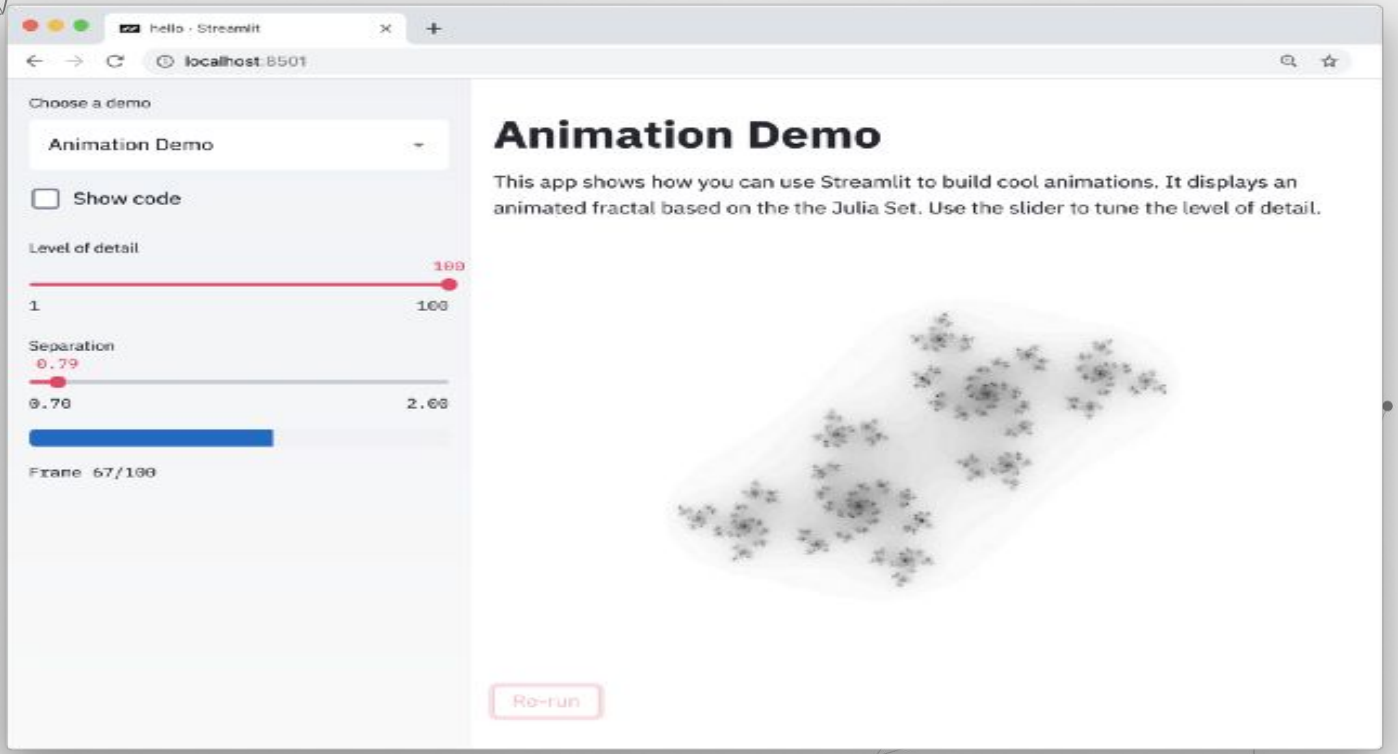


03

Install Streamlit & Play

```
$ pip install streamlit

$ streamlit hello
```



04

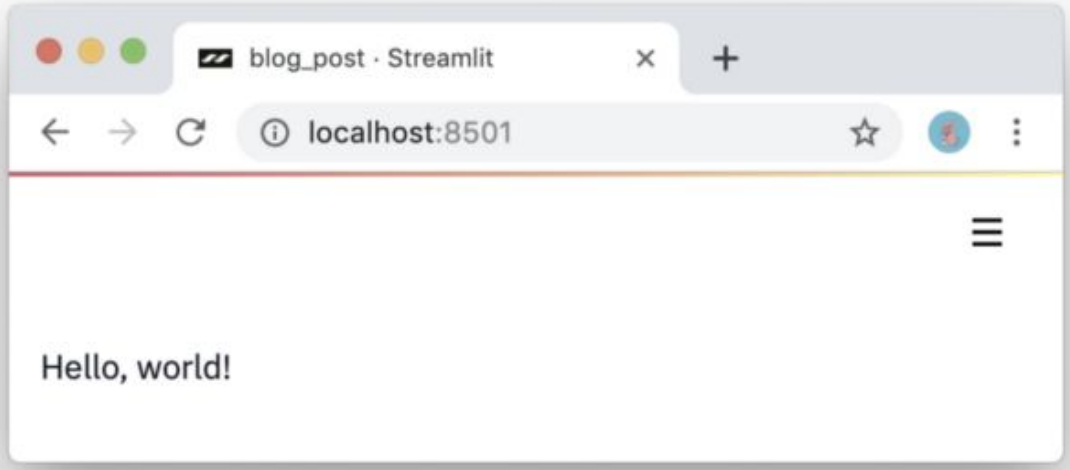
Building a data app

Streamlit process' walkthrough

```
import streamlit as st

st.write('hello world')
```

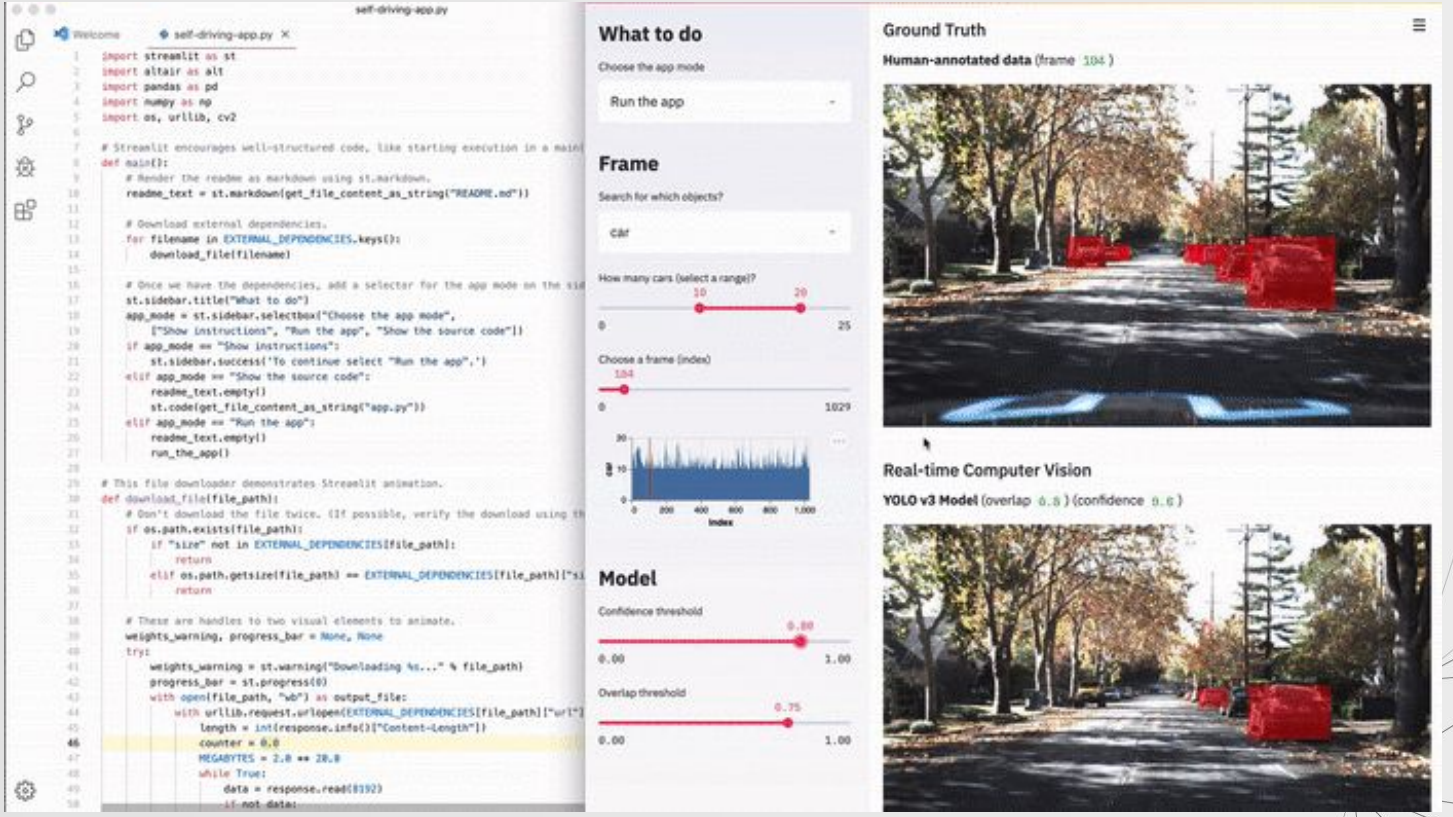
```
$ streamlit run hello.py
```



05

Sneak Peek & Use Cases

This Streamlit demo lets you perform semantic search across the **entire Udacity self-driving car photo dataset**, visualize human-annotated ground truth labels, and run a **complete neural net (YOLO)** in real time from within the app, in **less than 300 lines of code**.



06

Maybe Streamlit is not that awesome...

The Cons of Streamlit

- The Convenience-Flexibility Tradeoff
- Size of data input (~~50mb~~ 200mb max for data upload)
- ~~Limited~~ support for video/animation
- You can't build an app with functionalities like
  - ~~User Authentication~~
  - Newsletter subscription
  - User-to-user interaction

Even still...

- It’s simple enough to use for every Python programmer.
- It’s superb if you want an interactive data app quickly.

Fullstack Flask developer tools:

Flask and at least, JS, HTML, CSS & JSON

Fullstack Streamlit developer tools:

Streamlit (and maybe Docker)

Thank You!

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